

From VCF to RDF: RML-Based Conversion Approach for the Semantic Representation of Variant Data



Elias Crum^{1,2}, Bart Buelens², Gökhan Ertaylan², Ruben Taelman¹
¹IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium
²Flemish institute for Technological Research (VITO) Mol, Belgium

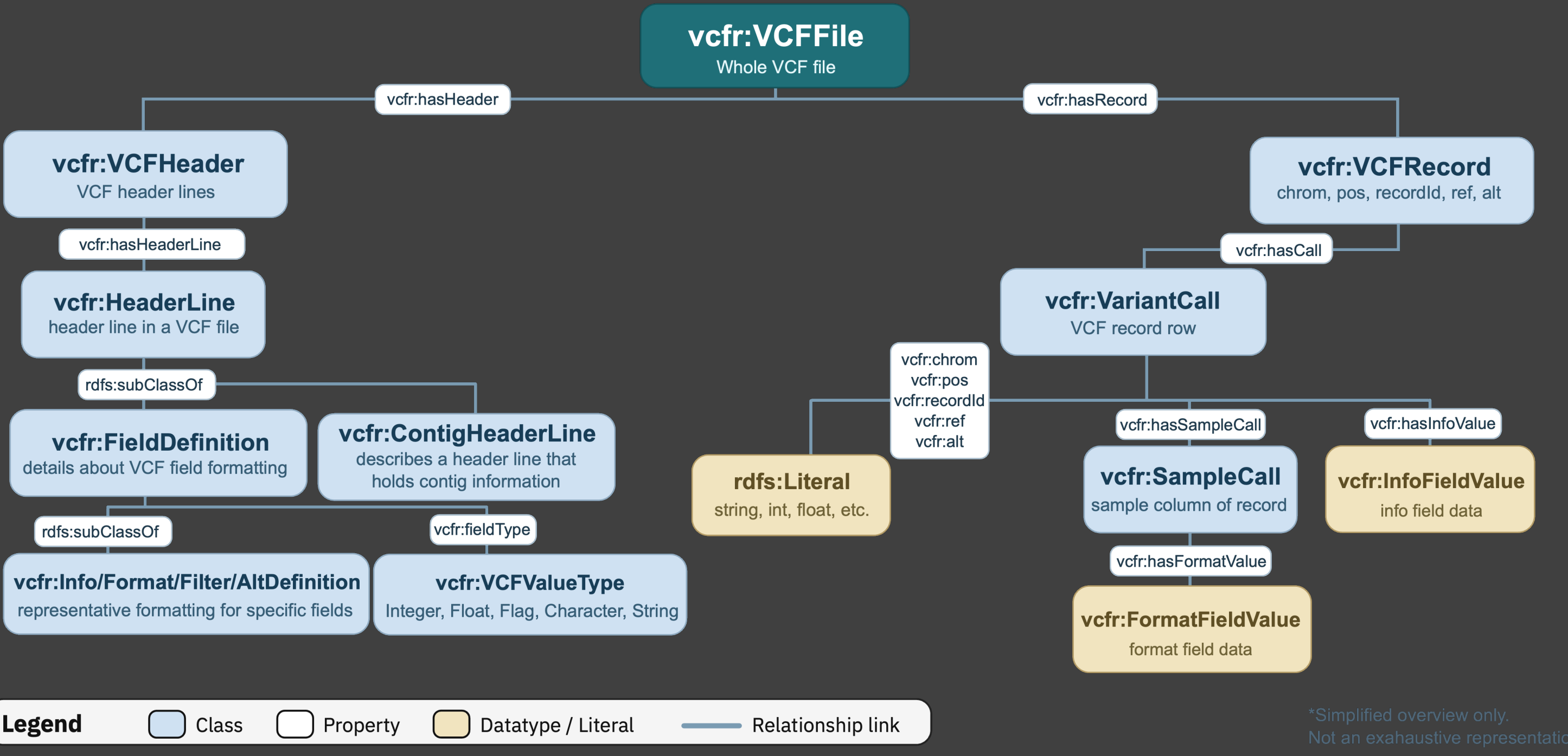


- Limited Query Capabilities**
 - Requires specialized tools
 - Not flexible or scalable
- Structural Complexity of VCF**
 - Highly heterogeneous and irregular fields:
 - INFO, FORMAT, GENOTYPE, HEADER
- Limited Interoperability**
 - Difficult to integrate with other biomedical datasets

Problem

VCF Vocabulary

- Challenges**
 - Weak Data Linking & Semantics**
 - Only inconsistent / weak references (e.g., rsIDs)
 - No rich semantic structure
 - Poor Data Access & Privacy Control**
 - All-or-nothing data sharing



- Semantic VCF Vocabulary**
 - <https://w3id.org/vcf-rdfizer/vocab#>
 - 1. Full VCF-to-RDF Representation
 - Models the entire VCF file
 - 2. Focus on Provenance & Structure
 - Explicitly captures HEADERS, row-level data, and call-level provenance
 - 3. Interoperability via Ontology Alignment
 - links to existing genomic ontologies (e.g., SO, FALDO, GENO, etc.)

RML for VCF Transformation

- 1. Declarative & Reusable Mappings
 - Define transformations as rules, not code.
- 2. Handles VCF complexity
 - Supports nested, irregular fields (INFO, FORMAT, GENOTYPE).
- 3. Semantic interoperability
 - Enables mapping to ontologies for FAIR, linked data.
- 4. Scales to large datasets
 - Streaming approach via RMLStreamer

RML Rule-Set

```
rml_rules.ttl

<#VCFRecordMap>
a rr:TriplesMap ;
rml:logicalSource [
  rml:source [
    a csvw:Table ;
    csvw:url "/data/tsv/records.tsv" ;
    csvw:dialect [
      a csvw:Dialect ;
      csvw:delimiter "\t" ;
      csvw:headerRowCount 1
    ] ;
  ] ;
  rml:referenceFormulation ql:CSV
] ;
rr:subjectMap [
  rr:template "file://{SOURCE_FILE}#record/{ROW_ID}" ;
  rr:class vcfr:VCFRecord
] ;
rr:predicateObjectMap [
  rr:predicate vcfr:chrom ;
  rr:objectMap [ rml:reference "CHROM" ]
] ;
...
```

+  **RMLStreamer Engine ¹**

¹ Gerald Haesendonck, et al. RMLio/RMLStreamer: V2.5.0. v2.5.0, Zenodo, 2 June 2023, <https://doi.org/10.5281/zenodo.7998156>.

VCF to RDF Example

```
example.vcf

##fileformat=VCFv4.2
##fileDate=20200615
##source=CLC Genomics Grid Worker 20.9.9 Beta 1 build 209900
##reference=/CLC_References/homo_sapiens/sequence/hg38_no_alt_analysis_set/Homo_sapiens_se
quence_hg38_no_alt_analysis_set
##contig=<ID=chr1,length=248956422>
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Total number of filtered reads used to
call variants at this position in this sample.">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allele depth, number of filtered reads
supporting the alleles, ordered as listed in REF and ALT.">

#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT SAMPLE1 SAMPLE2
chr1 123456 rs123 A T 52.3 PASS DP=42;AF=0.2 GT:DP:AD 0/1:42:30,12 0/1:6:3,2 ] ;
```

RDF Conversion

example.ttl#Header

```
@base <file://> .
@prefix vcfr: <https://w3id.org/vcf-rdfizer/vocab#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

<example-file1.vcf>
a vcfr:VCFFile ;
vcfr:fileFormat "VCFv4.2" ;
vcfr:fileDate "2020-06-15"^^xsd:date ;
vcfr:sourceSoftware "CLC Genomics Grid Worker
20.9.9 Beta 1 build 209900" ;
vcfr:referenceGenome
"/CLC_References/homo_sapiens/sequence/hg38_no_alt_an
alysis_set/Homo_sapiens_sequence_hg38_no_alt_analysis
_set" ;
vcfr:contigCount 1 ;
vcfr:headerKey <example-file1.vcf#header> ;
vcfr:hasRecord <example-file1.vcf#record/var1> .

<example-file1.vcf#header/line/1>
a vcfr:FileFormatHeaderLine ;
vcfr:headerKey "fileformat" ;
vcfr:headerValue "VCFv4.2" .

<example-file1.vcf#header/line/2>
a vcfr:FileDateHeaderLine ;
vcfr:headerKey "fileDate" ;
vcfr:headerValue "20200615" .
...
```

example.ttl#Record

```
@base <file://> .
@prefix vcfr: <https://w3id.org/vcf-rdfizer/vocab#> .

<example-file1.vcf>
a vcfr:VCFFile ;
vcfr:fileFormat "VCFv4.2" ;
vcfr:hasRecord <example-file1.vcf#record/var1> .

<example-file1.vcf#record/var1>
a vcfr:VCFRecord ;
vcfr:chrom "chr1" ;
vcfr:chromosome <example-file1.vcf#header/line/5> ;
vcfr:pos 123456 ;
vcfr:recordId "rs123" ;
vcfr:ref "A" ;
vcfr:alt "T" ;
vcfr:hasCall <example-file1.vcf#call/var1> .

<example-file1.vcf#call/var1>
a vcfr:VariantCall ;
vcfr:qual 52.3 ;
vcfr:filter "PASS" ;
vcfr:infoRaw "DP=42;AF=0.2" ;
vcfr:formatRaw "GT:DP:AD" ;
vcfr:hasInfoValue
  <example-file1.vcf#call/var1/info/AF>,
  <example-file1.vcf#call/var1/info/DP> ;
...
```



Future Work: • Testing of RML Mappings + Conversion Outputs • Vocabulary Improvements

